# Why `eos_widget` §12b Panel 1 goes negative and Panel 2 goes positive

A loss/function-surface explanation (it is *not* a bug)

## 1 What the two panels plot

The Edge-of-Stability tool's §12b panels 1 and 2 both plot, as a function of training step, the per-sample *product*

$$\text{product}_i \;=\; \big|\langle J_i,\, u\rangle\big| \;\cdot\; \sigma \;\cdot\; r_i, \tag{1}$$

averaged over the $N$ samples. Here $J_i = \nabla_\theta f_i$ is the parameter-gradient of the scalar ground-truth-label logit $f_i(\theta)$, $r_i$ is the per-sample residual, and $(\sigma, u)$ is an eigenpair of the **per-sample function Hessian** $Q_i = \nabla^2_\theta f_i$ — the Hessian of the scalar logit $f_i$ with respect to the network parameters $\theta$ (*not* the loss Hessian, and *not* the Gauss–Newton / GGN operator). **Panel 1** uses $u = u_{\text{top}}$, the *top* eigenvector of $Q_i$ (largest eigenvalue $\sigma_{\text{top}} > 0$, the direction of maximum upward curvature); **Panel 2** uses $u = u_{\text{bot}}$, the *bottom* eigenvector (most-negative eigenvalue $\sigma_{\text{bot}} < 0$, maximum downward curvature). Both panels multiply by the *same* residual $r_i$, and the magnitude factor $|\langle J_i, u\rangle| \geq 0$ is sign-free. The observation that Panel 1 drifts increasingly negative while Panel 2 drifts increasingly positive (both growing in magnitude) is exactly what the indefinite geometry of $Q_i$ predicts.

## 2 The function Hessian is indefinite — a saddle

The single fact that explains everything is that $\nabla^2 f_i$ is **indefinite**: it has both positive and negative eigenvalues. This is confirmed directly from the real per-sample spectra (Figure 1D): across 4 samples $\times$ 80 parameters = 320 eigenvalues, 44.7% **are negative** and 44.4% **are positive** (the remainder are numerically zero). The spectrum straddles zero, so $f_i(\theta)$, viewed as a function of the parameters near the current point, is a **saddle**, locally

$$f_i(\theta_0 + a\,u_{\text{top}} + b\,u_{\text{bot}}) \;\approx\; f_i(\theta_0) \;+\; \tfrac{1}{2}\big(\sigma_{\text{top}}\,a^2 + \sigma_{\text{bot}}\,b^2\big), \qquad \sigma_{\text{top}} > 0,\;\; \sigma_{\text{bot}} < 0. \tag{2}$$

Along $u_{\text{top}}$ the surface *curves up* ($\sigma_{\text{top}} > 0$, an upward parabola); along $u_{\text{bot}}$ it *curves down* ($\sigma_{\text{bot}} < 0$, a downward parabola). The real extremal eigenvalues are

$$\sigma_{\text{top}} \;\in\; [\,+1.64,\; +5.37\,], \qquad \sigma_{\text{bot}} \;\in\; [\,-11.03,\; -2.40\,]. \tag{3}$$

Note that $|\sigma_{\text{bot}}|$ can substantially exceed $|\sigma_{\text{top}}|$ ($11.03 > 5.37$); this asymmetry is why Panel 2's magnitude tends to *overshoot* Panel 1's.

## 3 The sign mechanism

Decompose Eq. (1) into its three factors:

$$\text{product}_i \;=\; \underbrace{\big|\langle J_i, u\rangle\big|}_{\geq 0 \text{ (sign-free)}} \;\cdot\; \underbrace{\sigma}_{\text{signed curvature}} \;\cdot\; \underbrace{r_i}_{\text{signed residual}}. \tag{4}$$

The first factor is an absolute value, hence non-negative. The residual is the *same* $r_i$ in both panels, and empirically the mean residual is **predominantly negative** ($r_i < 0$): the model under-predicts the ground-truth-label logit and is being pushed up. With $|\langle J_i, u\rangle| \geq 0$ and $r_i < 0$ fixed, the *only* factor that differs between the two panels is the *sign of* $\sigma$. Therefore

$$\text{Panel 1 (top):}\quad \big|\langle J_i, u_{\text{top}}\rangle\big| \cdot (+\sigma_{\text{top}}) \cdot (r_i < 0) \;\Longrightarrow\; \textbf{NEGATIVE}, \tag{5}$$

$$\text{Panel 2 (bottom):}\quad \big|\langle J_i, u_{\text{bot}}\rangle\big| \cdot (-|\sigma_{\text{bot}}|) \cdot (r_i < 0) \;\Longrightarrow\; \textbf{POSITIVE}. \tag{6}$$

The two panels are exact **mirror images**: the sign is inherited entirely from the curvature sign of the chosen eigenvector. The factor of $-1$ between $\sigma_{\text{top}} > 0$ and $\sigma_{\text{bot}} < 0$ propagates straight through Eq. (1).

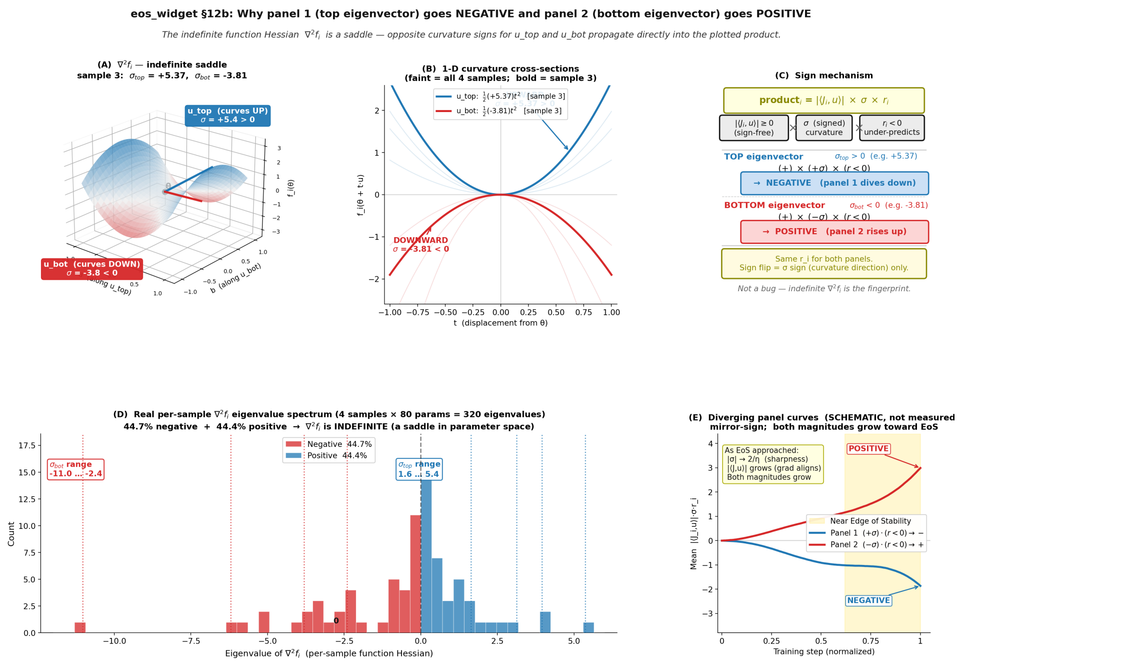


Figure 1: **The sign split is inherited from the indefinite function Hessian.** **(A)** The per-sample function surface rendered as the indefinite saddle $z = \frac{1}{2}(\sigma_{\text{top}} a^2 + \sigma_{\text{bot}} b^2)$ with real sample-3 curvatures ($\sigma_{\text{top}} = +5.37$, $\sigma_{\text{bot}} = -3.81$): the $u_{\text{top}}$ axis curves up ($\sigma > 0$), the $u_{\text{bot}}$ axis curves down ($\sigma < 0$). **(B)** 1-D cross-sections: an upward parabola along $u_{\text{top}}$ and a downward parabola along $u_{\text{bot}}$ (faint = all four samples, bold = sample 3). **(C)** The sign mechanism of Eq. (1): the magnitude factor $|\langle J_i, u\rangle| \geq 0$ is sign-free and the residual $r_i < 0$ is shared, so the only sign-carrier is $\sigma$. A $+\sigma$ (top) with $r < 0$ gives a *negative* product; a $-\sigma$ (bottom) with the same $r < 0$ gives a *positive* product. **(D)** The real per-sample $\nabla^2 f_i$ eigenvalue spectrum (320 eigenvalues): 44.7% negative, 44.4% positive — it straddles zero, hence indefinite. Dotted lines mark each sample's $\sigma_{\text{top}}$ (blue) and $\sigma_{\text{bot}}$ (red). **(E)** Schematic (not measured) of the two diverging panel curves: mirror-sign, both magnitudes growing as the Edge of Stability is approached.

# 4 Why both magnitudes grow toward the Edge of Stability

The user's intuition that "both should increase" is correct *for the magnitude*; only the sign differs. As training approaches the Edge of Stability, two effects compound: (i) **progressive sharpening** grows the extremal curvatures, $|\sigma| \to 2/\eta$ (the sharpness ceiling set by the learning rate $\eta$); and (ii) **gradient alignment** rotates $J_i$ toward the extremal eigenvectors, so $|\langle J_i, u\rangle|$ grows for both $u_{\text{top}}$ and $u_{\text{bot}}$. Both effects raise $|\text{product}_i|$ in *both* panels, so Panel 1 dives *more* negative and Panel 2 climbs *more* positive (Figure 1E) — diverging in sign but growing together in magnitude. Because $|\sigma_{\text{bot}}| > |\sigma_{\text{top}}|$, the bottom panel typically grows faster.

# 5 Conclusion: not a bug

The opposite directions of the two panels are the **fingerprint of the indefinite function Hessian** (a saddle in parameter space), not an error. The chain is:

$$\begin{aligned} \nabla^2 f_i \text{ indefinite } &\Rightarrow \sigma_{\text{top}} > 0,\ \sigma_{\text{bot}} < 0 \\ &\Rightarrow \text{shared } r_i < 0 \text{ flips sign with } \sigma \\ &\Rightarrow \text{Panel 1 negative, Panel 2 positive.} \end{aligned}$$

For contrast, the **Gauss–Newton / GGN** operator is positive-semidefinite ($\sigma \geq 0$): if the tool tracked it instead, both "top" and "bottom" directions would carry a non-negative $\sigma$ and the two panels *would share a sign.* The observed mirror-image behavior is thus a direct, expected consequence of choosing the indefinite $\nabla^2 f_i$ as the curvature operator.